

balance and other edits to your photo before you even create the photo. Photo editing applications such as Photoshop handle most exposure and white balance editing on JPEG, TIFF and Photoshop files by removing data from the photo. To test this out, launch a photo in Photoshop, and open the Levels panel (**Image > Adjustments > Levels**) where you'll see a histogram for your photo. At the bottom of the histogram will be three diamond-shaped sliders. Move in the left and right sliders toward the centre. In most photos, this will have the effect of increasing the contrast of your photo. Click OK, and your levels changes will be accepted. Then reopen the Levels panel, and you'll most likely see the histogram with open vertical lines. These lines indicate missing information or data that has been removed from the photo in order to achieve the effect you saw on your monitor.

With a RAW photo, you can make these levels and other adjustments before you use the conversion algorithms. In Photoshop CS, this is done when you double-click on a RAW photo. Photoshop will present you with a panel with various exposure, white balance, contrast, saturation, shadow and other adjustment options in addition to a very large thumbnail of your photo. You can play with these adjustments and Photoshop will adjust the thumbnail photo to give you a very good representation of what the final photo will look like if you accept these settings. When you click OK, your editing adjustments are made to the algorithms that convert your RAW data to reflect the changes you want to make in the photo.

The most important thing to remember here is that your adjustments are being applied while your photo is being created. This means that no data is lost, and your photo will have all the tonal characteristics that are possible with that file. Because no data is lost when

you adjust a RAW file, this gives you the ability to make exposure changes to your photo that are as wide as two f-stops or more. This is the reason many commercial photographers shoot in RAW.

Lenses For Digital Cameras

If you're planning to buy a lens for your digicam, how do you distinguish between the different kinds of lenses? Wide-angle or telephoto? What focal length and what speed? There are several questions you need to answer for yourself. Here's a brief primer.

Choosing The Focal Length

Focal length is probably the most important factor to be considered when choosing a lens: it determines the field-of-view of the photos you will be able to take with your camera.

The two main types of focal length are telephoto and wide-angle, and while telephoto lenses have a narrow field-of-view and are best suited for close-up shots and portraits, wide-angle lenses have a wider field-of-view which is perfect for indoor photography and landscapes.

Keep in mind that the performance of lenses differs from camera to camera.

Fast And Slow

When you hear about fast and slow lenses, people are referring to a lens' maximum aperture, which is the maximum amount of light that a lens can let in. A simple rule of thumb is that a fast lens lets in a lot of light, while a slow lens lets in less light, which defines how your photos will look.

Maximum apertures are measured in f-stop numbers, which are actually a ratio of the size of the lens aperture and focal length. The smaller the f-stop number, the more light is let in. An increment in the f-stop number doubles the amount of light let in, so f1.4 lets more light than f2.0.

This may seem confusing at first; the easiest way to make sense of it is to remember that fast lenses are best suited towards successful photography in darker lighting conditions, and slow lenses are targeted towards photography in brighter conditions.

Zoom Lenses

Unlike a fixed-focal-length lens, a zoom lens often gives you the diversity of a range of focal lengths all rolled into a single adjustable lens. This can be great if you often have to switch between various lenses for different shots, but it is important to remember that not all zoom lenses have a constant maximum aperture, and those that do are often larger and more expensive.

The maximum aperture can be reduced as you zoom in using a lens with a variable maximum aperture, but this

may not be as important to some as the reduced cost and size of such lenses.

Add-on Lenses

Add-on or accessory lenses are targeted at compact digital cameras, and allow you to significantly lengthen or reduce the camera's inbuilt focal length while at the same time being able to automate camera functions including f-stop settings and focusing. These lenses can be excellent low-cost add-ons to your digital camera, with telephoto add-on lenses being able to increase focal lengths by up to 300 per cent, and wide-angle versions allowing for reduction in focal lengths of up to 30 per cent.

A fish-eye lens is an extreme wide-angle lens that has an angle of view exceeding 100 degrees—sometimes more than 180 degrees—and which renders a scene as highly distorted. ■



Above: A photograph that's clicked normally

Below: The same scene shot using a fish-eye lens

